

A related *consumption* explanation is that the real rate is a price that investors use to gauge how expensive it is to shift consumption today to the future. It reflects the cost of foregoing eating today and instead eating tomorrow. Usually, consumption today is preferred, so eating one fruit today is equivalent to eating 1.02 fruits next year. But with a negative real rate, consumers prefer to save: eating tomorrow is actually better than eating today. This reflects extreme *precautionary saving*. Consumers are willing to shift consumption to later from today because they believe very bad times are a-coming—taxes will be high, economic growth will be low, there will be even more undue regulatory burdens, and even greater political uncertainty than today.

Periods of negative real rates are not uncommon. In Ang, Bekaert, and Wei (2008), I estimate real yields since 1952. Even though real yields were not traded in the United States prior to 1997, I can estimate real rates using a term structure model (see chapter 9), which ties together inflation and nominal yields, with assumptions on time-varying risk premiums, to infer the term structure of real rates. I find pronounced negative short-term real rates in the late 1950s and a second period extending over the late 1970s and early 1980s. Both periods were associated with accelerating inflation. I also find very long-maturity real rates have not turned negative. This is consistent with the TIPS data in the more recent sample where the twenty-year TIPS yield has not turned negative: on December 2011, the twenty-year TIP had a yield of 0.53% even though the five-year TIP had a negative yield.

The variability of real rates is one reason TIPS are a weak hedge against inflation. Over the sample for the data in Figure 11.5, the correlations of five-, ten-, and twenty-year TIPS yields with inflation are negative, at -23%, -12%, and -13%, respectively. As inflation increases, real yields tend to fall: unfortunately at the time investors most desire high real yields—when inflation is high—they are not forthcoming. Due to time-varying real rates, real bond portfolios are not so real.

4.4. BREAK-EVEN INFLATION COMPENSATION

In Figure 11.5, Panels B through D, Treasuries always have higher yields than TIPS and the distance between Treasuries and TIPS is fairly constant, except for the 2008 and 2009 financial crisis. The difference between nominal and real bond yields is labeled break-even inflation or inflation compensation (I use the two terms interchangeably, as the literature does):

$$\begin{aligned} \text{Nominal yield} &= \text{Real yield} + \text{Break-Even Inflation} \\ \text{or Nominal yield} &= \text{Real yield} + \text{Inflation Compensation} \end{aligned} \tag{11.1}$$